

UMC 0.11um RFCMOS Al Advanced Enhancement Resistor SPICE Model Document

**Version 0.2 Phase 1
2010/07/23**

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1 Introduction

1.1 Revision history

Table 1-1 Revision history

Version	Date	Author	Remark
0.1_p1	2010/02/05	Ian Chen	1. Original release.
0.1_p2	2010/04/12	Shih-Hsin Yeh	1.Revise Rend, Rsh, tcr1, tcr2, vcr1, vcr2, and dw based on EDR.
0.2_p1	2010/07/23	Yealing Lin	1.Modified Non-Salicided P ⁺ Poly RF resistor and Non-Salicided N ⁺ Poly RF resistor model based on new silicon data. 2.Non-Salicided HR Poly RF resistor model original released.

2 RF Model

2.1 General Information

This document describes poly resistor RF models for UMC's 110nm Logic and Mixex-Mode Mteal Metal Capacitor AI Advanced Enhancement process.

The voltage and temperature responses for the RF resistor models are based on silicon characterization data.

The high-frequency responses for the RF resistors models in this document are generated based on microwave scattering-parameter (S-parameter) simulation.

In current release, 3 major commercial simulators (Cadence's SPECTRE, Synopsys' HSPICE and Mentor Graphics' ELDO) are supported.

It is suggested to read the model document first to understand the model usage and limitation before using the models.

2.2 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information

	Non-Salicated P ⁺ Poly Resistor	Non-Salicated N ⁺ Poly Resistor	Non-Salicated HR Poly Resistor
Mask ID	BSW0FA	BSW0FA	BSW0FA
Lot ID	D4YAA	D4YAA	D4YAA
Wafer Number	#01	#01	#01

2.3 Non-salicated P+ Poly Resistor RF Sub-circuit Model

Model File Name:

Spectre: L110AE_RNPPO_RF_V021.lib.scs
L110AE_RNPPO_RF_V021.mdl.scs

Hspice: L110AE_RNPPO_RF_V021.lib
L110AE_RNPPO_RF_V021.mdl

Eldo: L110AE_RNPPO_RF_V021.lib.eldo
L110AE_RNPPO_RF_V021.mdl.eldo

Inline Sub-circuit Model Name & Node Definition:

rnppo_rf (port1 port2 nwell psub) -- nonsalicyded P+ poly resistor over nwell

Corner Section:

- tt : typical R value
- ff : minimum R value
- ss : maximum R value

Input Parameters:

- l: length in m
- w: width in m
- m: multiplier factor (default is 1)
- dx: distance between resistor edge to nwell edge in l direction, default is 1.68u (for rnppo_rf only)
- dy: distance between resistor edge to nwell edge in w direction, default is 1.68u (for rnppo_rf only)

Simulator Versions:

Synopsys HSPICE release 2007.09

Cadence SPECTRE version 6.2.1.299

Mentor Graphics ELDO version 6.7_1.1

Model Usage:

1. For Cadence Spectre

- Copy model files into your simulation directory
- Add the following statement in the input netlist to the simulator:
include "/ L110AE_RNPPO_RF_V021.lib.scs" section=CornerSection
The CornerSection could be tt, ss or ff
- Follow the following example to call the model:
x1 (port1 port2 nwell psub) rnppo_rf l=20e-6 w=2e-6 m=1

2. For Synopsys Hspice

- Copy model files into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_RNPPO_RF_V021.lib " CornerSection
, where CornerSection could be tt, ss, ff.
- Follow the example below to call the model
x1 port1 port2 nwell psub rnppo_rf l=20e-6 w=2e-6 m=1

3. For Mentor Graphics Eldo

- Copy model files into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_RNPPO_RF_V021.lib.eldo " CornerSection
, where CornerSection could be tt, ss, ff.
- Follow the example below to call the model
x1 port1 port2 nwell psub rnppo_rf l=20e-6 w=2e-6 m=1

Scope:

In this report, the RF modeling results of nonsalicide P+ poly resistor are described.

1. Device Architecture and Layout

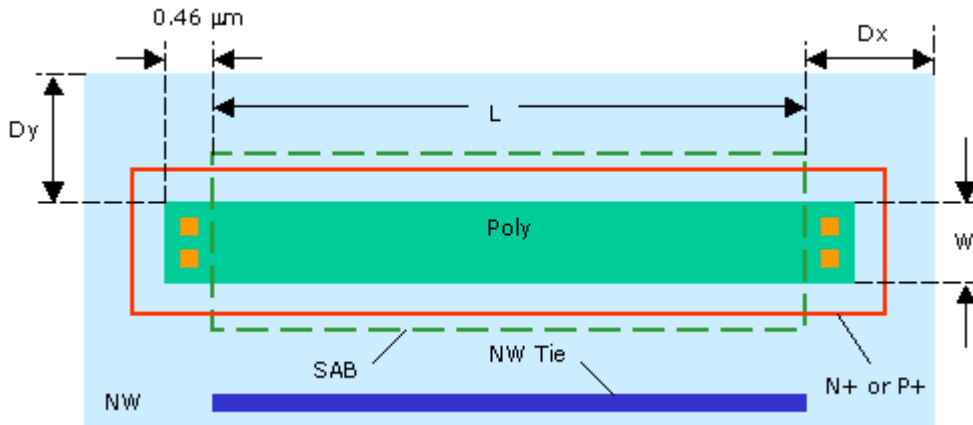


Figure 2-1 Example layout for nonsalicided P+ poly RF resistor

2. Extension equivalent circuit model

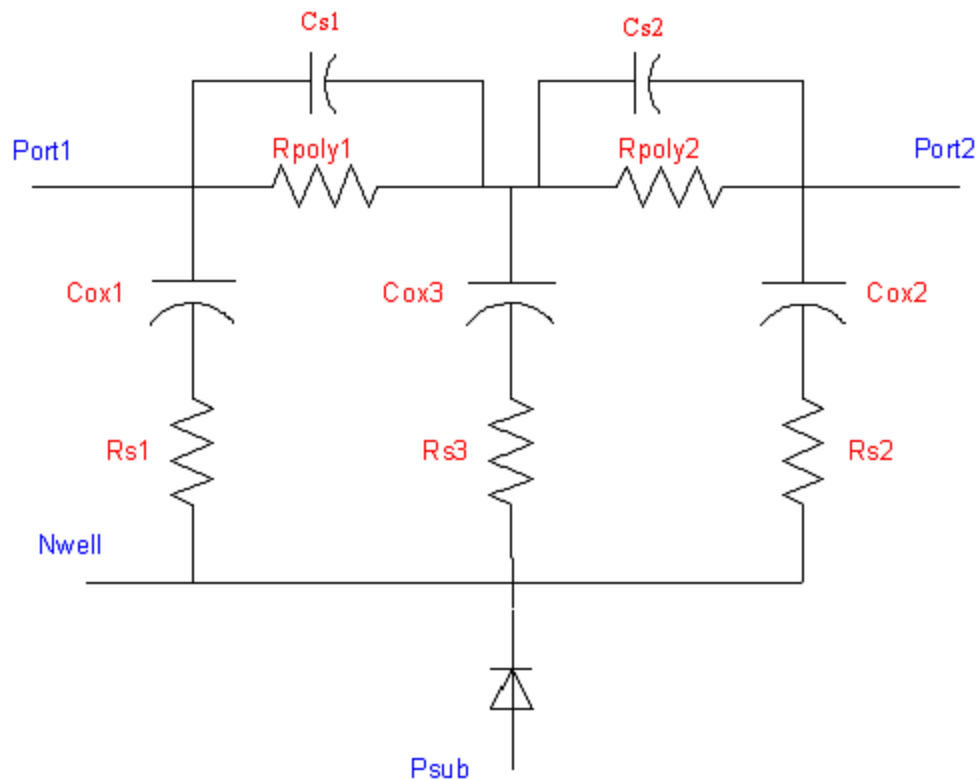


Figure 2-2 Sub-Circuit of RF P+ poly resistors

3. Recommended Scaling Range

Geometry scaling:

- Width (w): 0.5um ~ 2um
- Length (l): 5 ~ 20um
- Frequency range: 100MHz ~ 20.1GHz
- Temperature range: -50°C ~ 125°C

4. Comparison between model & measurement over frequency

*.m: measurement data

*.s: simulation data

4.1 DC & RF Fitting curves for individual devices

Effective series resistance: $R_p = -\text{Re}[1/Y_{12}]$

Effective shunt capacitance: $C_{ox} = -1/\{\omega \cdot \text{Im}[1/(Y_{11}+Y_{12})]\}$, ω is angular frequency.

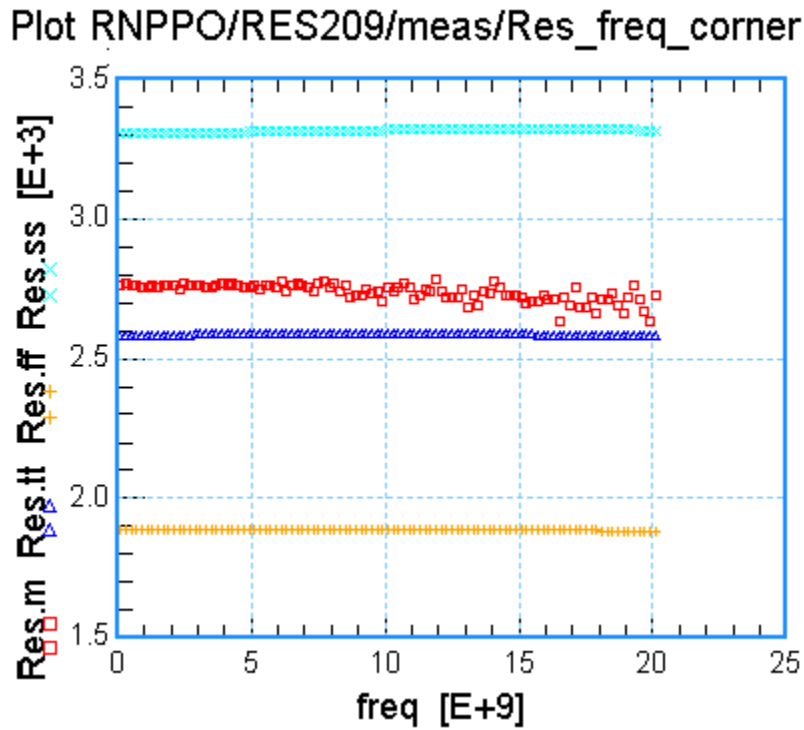


Figure 2-3 Frequency response of Rs for W=1um, L=10um, m=1 P+ poly resistor

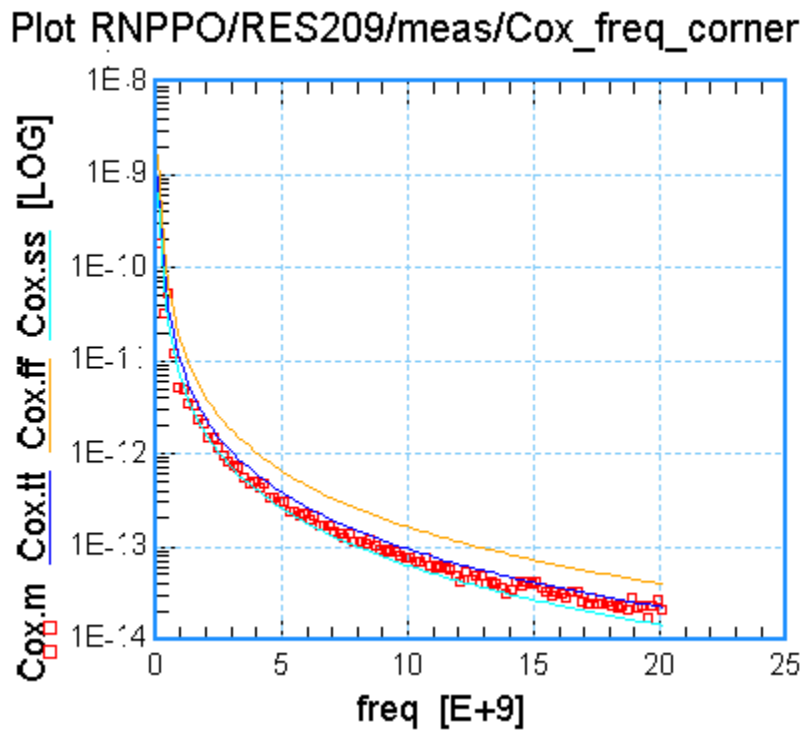


Figure 2-4 Frequency response of Cox for W=1um, L=10um, m=1 P+ poly resistor

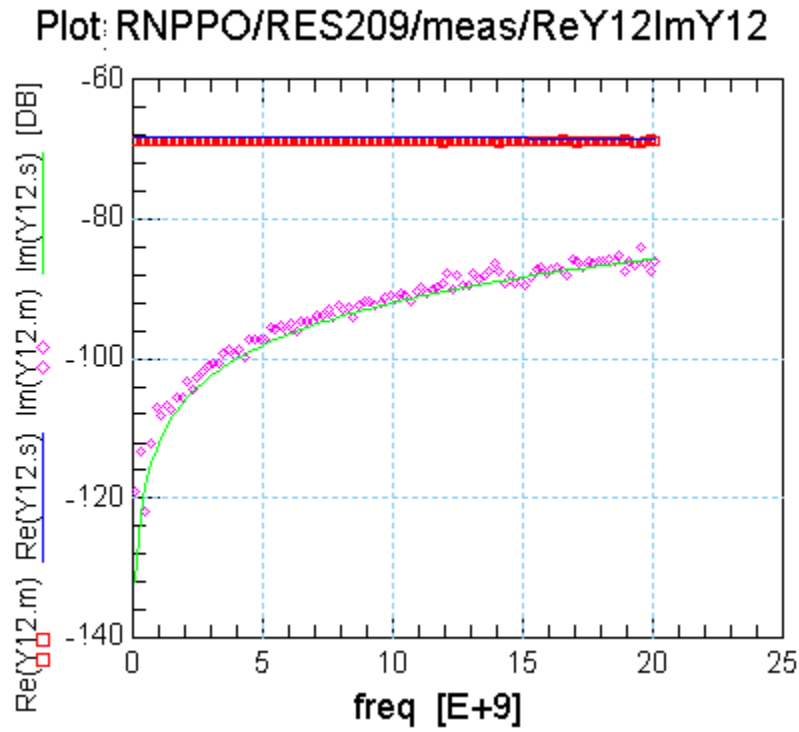


Figure 2-5 Frequency response of Re(Y12) & Im(Y12) vs. Frequency for W=1um, L=10um, m=1 P+ poly resistor

2.4 Non-salicided N+ Poly Resistor RF Sub-circuit Model

Model File Name:

Spectre: L110AE_RNNPO_RF_V021.lib.scs
L110AE_RNNPO_RF_V021.mdl.scs

Hspice: L110AE_RNNPO_RF_V021.lib
L110AE_RNNPO_RF_V021.mdl

Eldo: L110AE_RNNPO_RF_V021.lib.eldo
L110AE_RNNPO_RF_V021.mdl.eldo

Inline Sub-circuit Model Name & Node Definition:

rnpno_rf (port1 port2 nwell psub) -- nonsalicided N+ poly resistor over nwell

Corner Section:

- tt : typical R value
- ff : minimum R value
- ss : maximum R value

Input Parameters:

- l: length in m
- w: width in m
- m: multiplier factor (default is 1)
- dx: distance between resistor edge to nwell edge in l direction, default is 1.68u (for rnpno_rf only)
- dy: distance between resistor edge to nwell edge in w direction, default is 1.68u (for rnpno_rf only)

Simulator Versions:

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.7_1.1

Model Usage:

1. For Cadence Spectre

- Copy model files into your simulation directory
- Add the following statement in the input netlist to the simulator:
include "./ L110AE_RNNPO_RF_V021.lib.scs" section=CornerSection
The CornerSection could be tt, ss or ff
- Follow the following example to call the model:
x1 (port1 port2 nwell psub) rnpno_rf l=20e-6 w=2e-6 m=1

2. For Synopsis Hspice

- Copy model files into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_RNNPO_RF_V021.lib " CornerSection
, where CornerSection could be tt, ss, ff.
- Follow the example below to call the model
x1 port1 port2 nwell psub rnnpo_rf l=20e-6 w=2e-6 m=1

3. For Mentor Graphics Eldo

- Copy model files into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_RNNPO_RF_V021.lib.eldo " CornerSection
, where CornerSection could be tt, ss, ff.
- Follow the example below to call the model
x1 port1 port2 nwell psub rnnpo_rf l=20e-6 w=2e-6 m=1

Scope:

In this report, the RF modeling results of nonsalicide P+ poly resistor are described.

1. Device Architecture and Layout

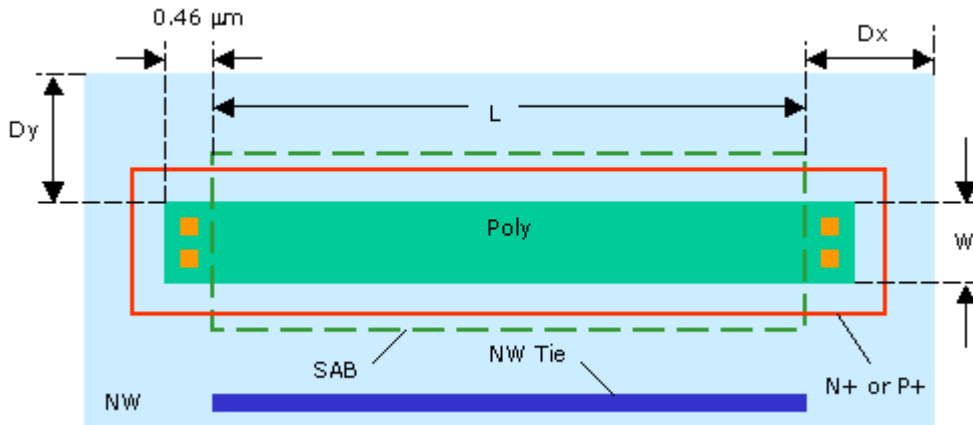


Figure 2-6 Example layout for nonsalicided N+ poly RF resistor

2. Extension equivalent circuit model

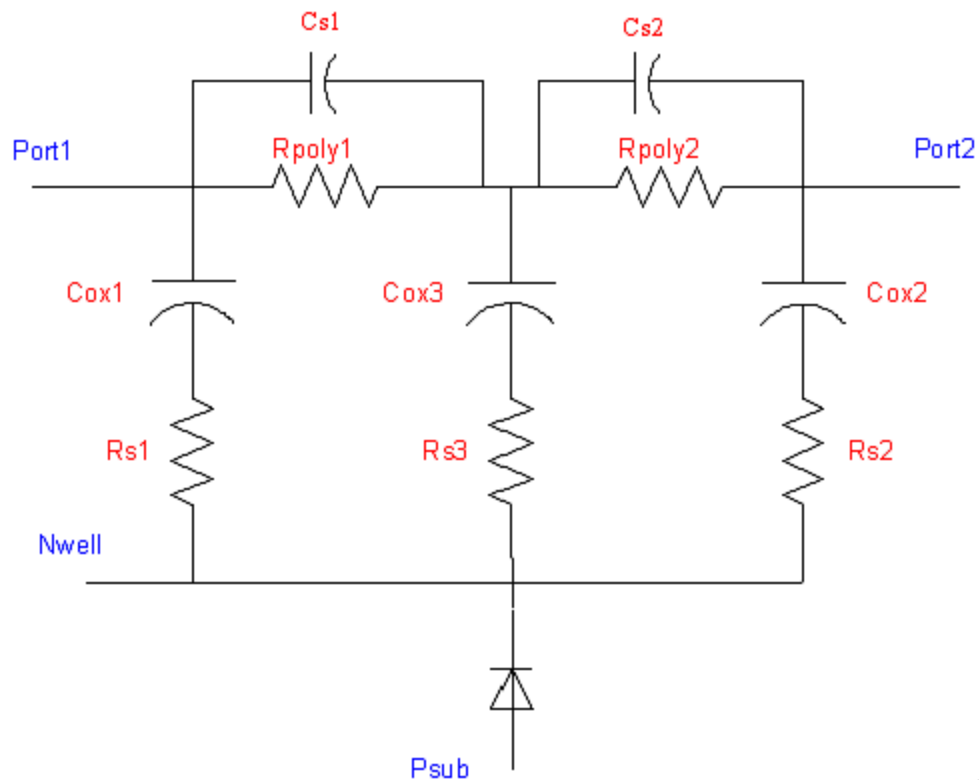


Figure 2-7 Sub-Circuit of RF N+ poly resistors

3. Recommended Scaling Range

Geometry scaling:

- Width (w): 0.5um ~ 2um
- Length (l): 5 ~ 20um
- Frequency range: 100MHz ~ 20.1GHz
- Temperature range: -50°C ~ 125°C

4. Comparison between model & measurement over frequency

*.m: measurement data

*.s: simulation data

4.1 DC & RF Fitting curves for individual devices

Effective series resistance: $R_p = -\text{Re}[1/Y_{12}]$

Effective shunt capacitance: $C_{ox} = -1/\{\omega \cdot \text{Im}[1/(Y_{11}+Y_{12})]\}$, ω is angular frequency.

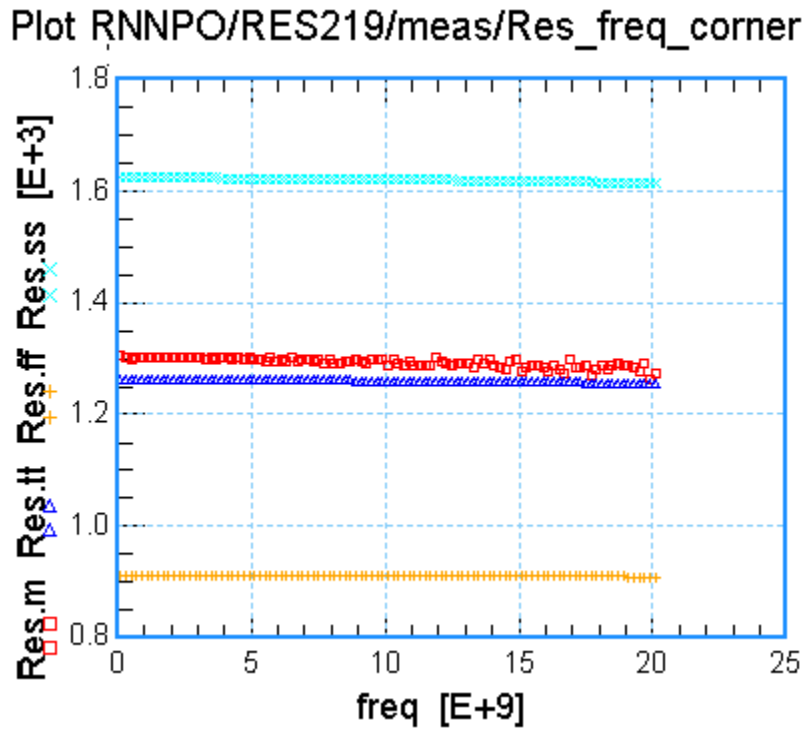


Figure 2-8 Frequency response of Rs for W=1um, L=10um, m=1 N+ poly resistor

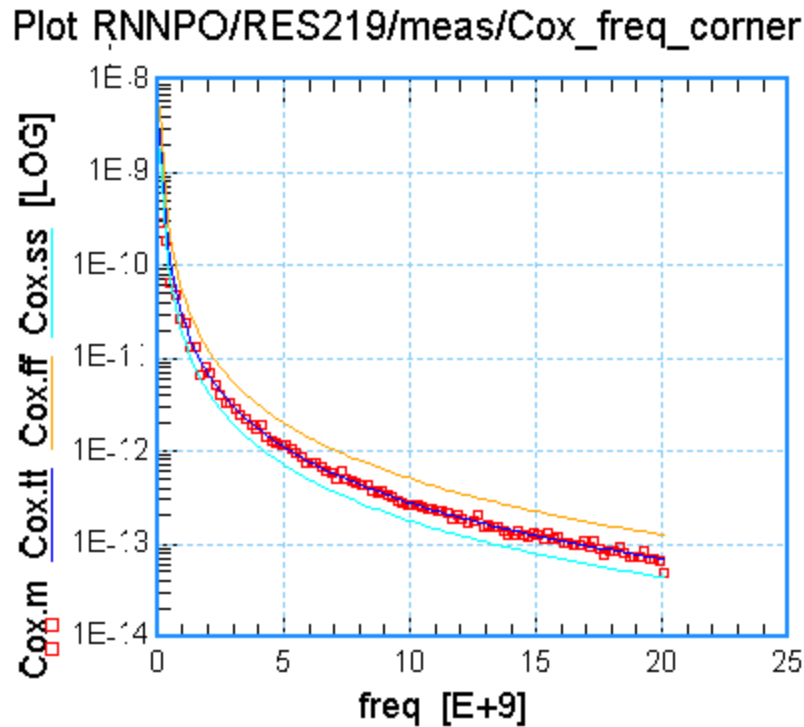


Figure 2-9 Frequency response of Cox for W=1um, L=10um, m=1 N+ poly resistor

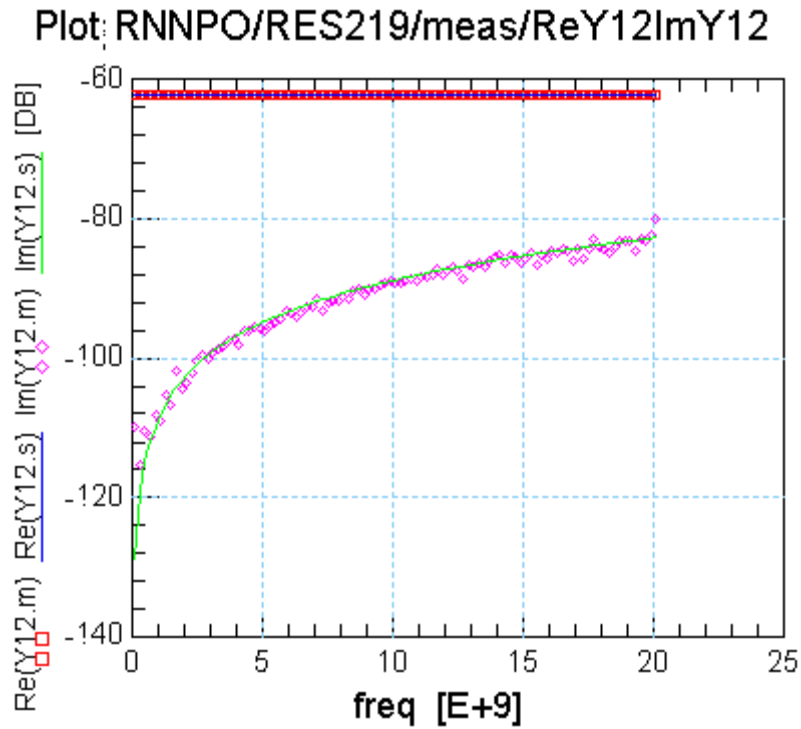


Figure 2-10 Frequency response of $\text{Re}(Y_{12})$ & $\text{Im}(Y_{12})$ vs. Frequency for $W=1\mu\text{m}$, $L=10\mu\text{m}$, $m=1$ N+ poly resistor

2.5 Non-salicided HR Poly Resistor RF Sub-circuit Model

Model File Name:

Spectre: L110AE_RNHR_RF_V021.lib.scs
L110AE_RNHR_RF_V021.mdl.scs

Hspice: L110AE_RNHR_RF_V021.lib
L110AE_RNHR_RF_V021.mdl

Eldo: L110AE_RNHR_RF_V021.lib.eldo
L110AE_RNHR_RF_V021.mdl.eldo

Inline Sub-circuit Model Name & Node Definition:

rnppo_rf (port1 port2 nwell psub) -- nonsalicided HR poly resistor over nwell

Corner Section:

- tt : typical R value
- ff : minimum R value
- ss : maximum R value

Input Parameters:

- l: length in m
- w: width in m
- m: multiplier factor (default is 1)
- dx: distance between resistor edge to nwell edge in l direction, default is 1.68u (for rnhr1000_rf only)
- dy: distance between resistor edge to nwell edge in w direction, default is 1.68u (for rnhr1000_rf only)

Simulator Versions:

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.7_1.1

Model Usage:

1. For Cadence Spectre

- Copy model files into your simulation directory
- Add the following statement in the input netlist to the simulator:
include "./ L110AE_RNHR_RF_V021.lib.scs" section=CornerSection
The CornerSection could be tt, ss or ff
- Follow the following example to call the model:
x1 (port1 port2 nwell psub) rnhr1000_rf l=20e-6 w=2e-6 m=1

2. For Synopsis Hspice

- Copy model files into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_RNHR_RF_V021.lib " CornerSection
, where CornerSection could be tt, ss, ff.
- Follow the example below to call the model
x1 port1 port2 nwell psub rnhr1000_rf l=20e-6 w=2e-6 m=1

3. For Mentor Graphics Eldo

- Copy model files into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_RNHR_RF_V021.lib.eldo " CornerSection
, where CornerSection could be tt, ss, ff.
- Follow the example below to call the model
x1 port1 port2 nwell psub rnhr1000_rf l=20e-6 w=2e-6 m=1

Scope:

In this report, the RF modeling results of nonsalicide HR poly resistor are described.

1. Device Architecture and Layout

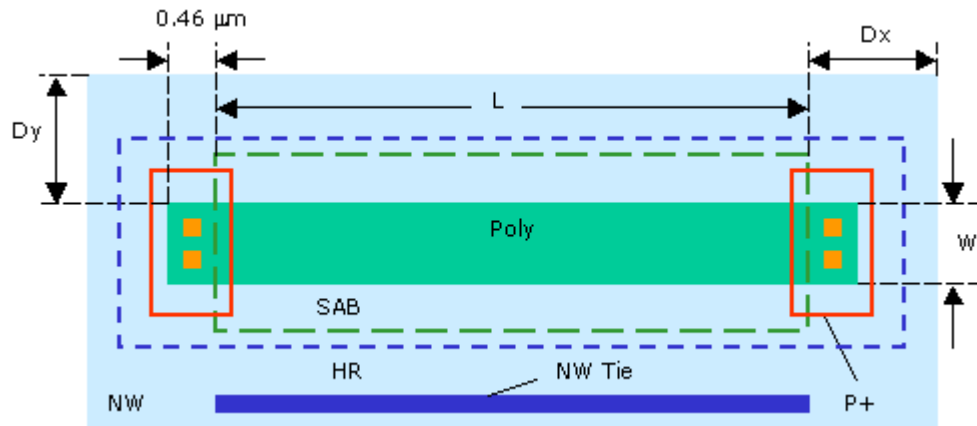


Figure 2-11 Example layout for nonsalicide HR poly RF resistor

2. Extension equivalent circuit model

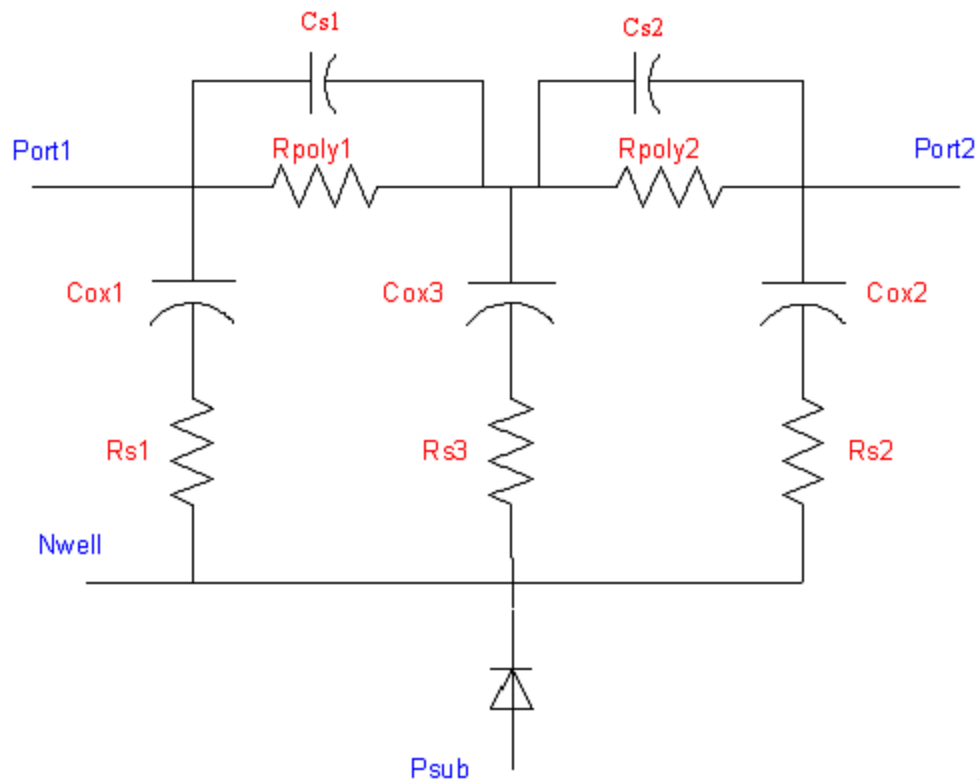


Figure 2-12 Sub-Circuit of RF HR poly resistors

3. Recommended Scaling Range

Geometry scaling:

- Width (w): 0.5um ~ 2um
- Length (l): 5 ~ 20um
- Frequency range: 100MHz ~ 20.1GHz
- Temperature range: -50°C ~ 125°C

4. Comparison between model & measurement over frequency

*.m: measurement data

*.s: simulation data

4.1 DC & RF Fitting curves for individual devices

Effective series resistance: $R_p = -\text{Re}[1/Y_{12}]$

Effective shunt capacitance: $C_{ox} = -1/\{\omega \cdot \text{Im}[1/(Y_{11}+Y_{12})]\}$, ω is angular frequency.

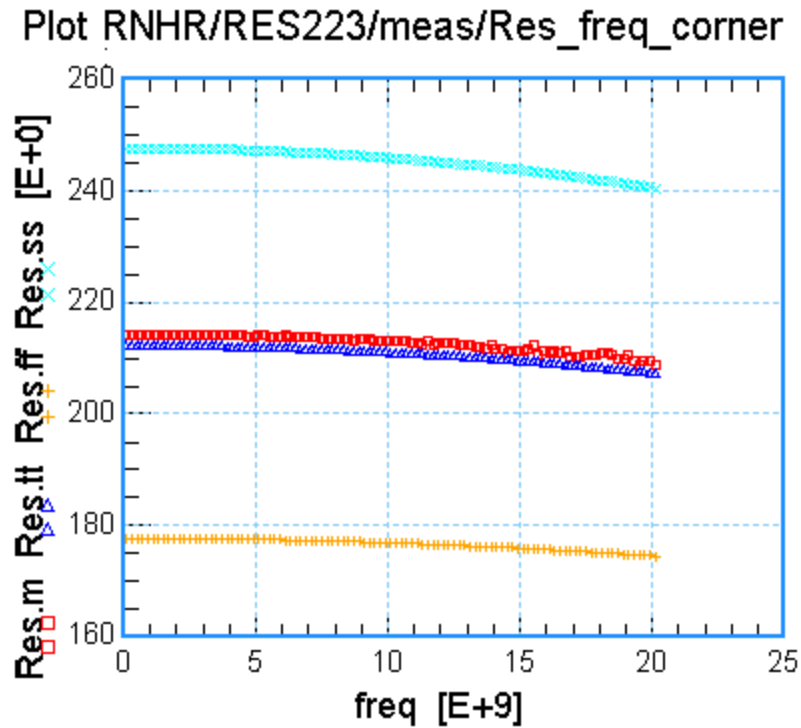


Figure 2-13 Frequency response of Rs for W=2um, L=8um, m=18 HR poly resistor

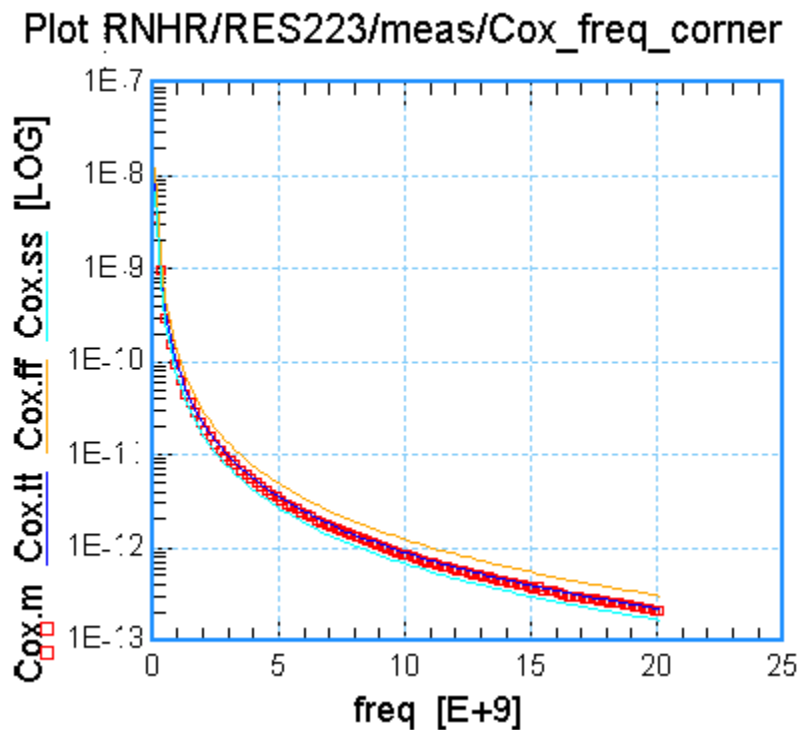


Figure 2-14 Frequency response of Cox for W=2um, L=8um, m=18 HR poly resistor

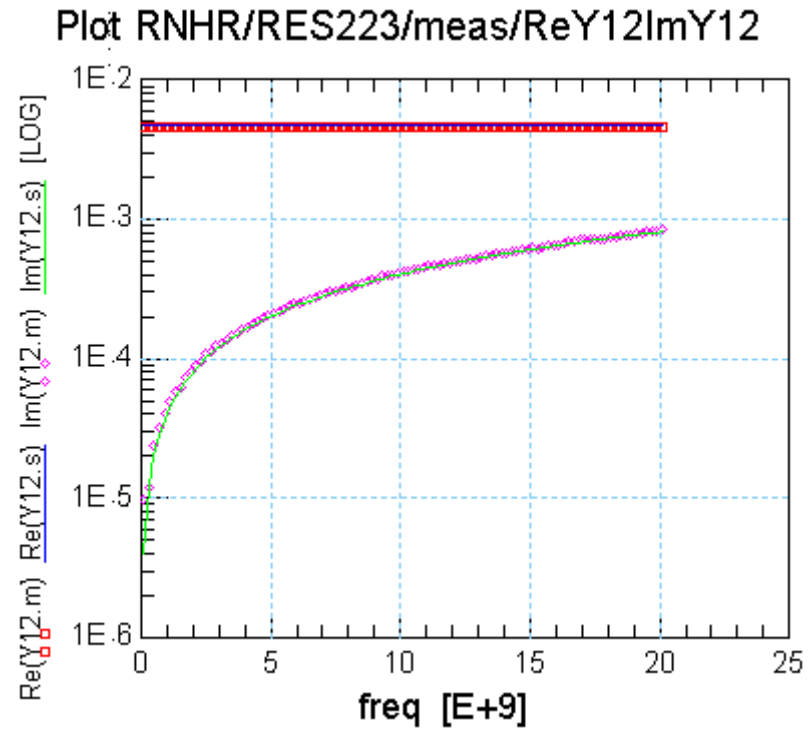


Figure 2-15 Frequency response of Re(Y12) & Im(Y12) vs. Frequency for W=2um, L=8um, m=18 HR poly resistor

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

Item	Warning Message	Explanation
1.	None	None

3.2 ELDO Warning Message

Table 3-2 ELDO warning message

Item	Warning Message	Explanation
1.	None	None

3.3 SPECTRE Warning Message

Table 3-3 SPECTRE warning message

Item	Warning Message	Explanation
1.	None	None